Development of Cheap Acoustic Modems and Communication Protocols for a Swarm of Underwater Manta Ray Robots

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ABSTRACT

Currently, development of underwater robotic swarms is inhibited by the lack of low-cost, reliable inter-robot communication methods. Acoustic communication has long been seen as the 'gold standard' for underwater communication; however, sensors are extremely expensive and thus not feasible for use in swarm robotic applications, where you need multiple sensors. Other methods, such as optical communication, require communicating robots to be in the same line of sight, or are unreliable in different lighting conditions. In this study, we aim to build low-cost inter-robot acoustic communication devices and protocols to allow a manta-ray robotic swarm to communicate with one another with the end goal of deploying these robots to map the ocean floor. Initially we have focused on developing a 3D printable modular manta-ray robot and prototyping acoustic circuits which utilize low cost 20mm piezoelectric transducers. Next steps will be to design communication protocols utilizing these transducers and to test them, first individually, then on locomoting robots. Once that testing is completed, we plan for these robots to be equipped with sonars to perform underwater mapping and ocean monitoring tasks in the Pamlico sound area.

APPLICATIONS

Underwater Mapping

- About 80% of the ocean floor is unmapped; an underwater robot swarm would make mapping an entire ocean/sea region faster.
- Low-cost acoustic provides affordable scaling of swarm which allows for faster mapping of ocean regions and a cheaper alternative for underwater communication.

Overall Ocean Monitoring

- The field of oceanography can be limited by cost, the Manta Ray Robot features cheap acoustic sensors.
- Our robots can be equipped with other sensors like water quality monitors, temperature probes, and salinity sensors for environment monitoring.

Bio-inspired Robotics

- The Manta Ray robot shape allows for efficient locomotion because the Manta Ray has the most body length per flap of it fins.
- The structure allows for efficient sonar placement and it is designed to keep the internal components fully waterproof.

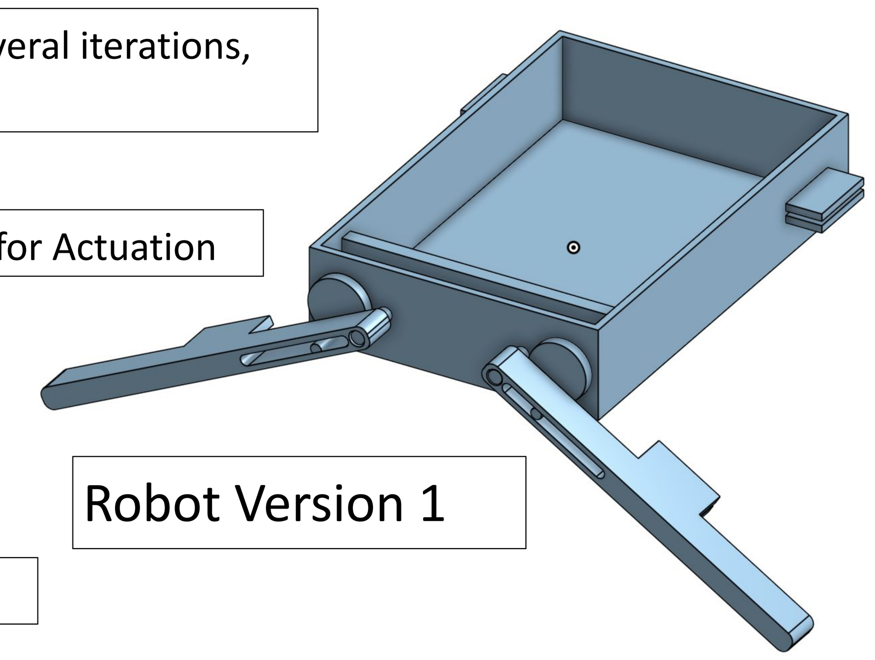
Iteration Overview

- A large portion of the design inspiration for this robot is making it waterproof cheaply, and keeping it a modular, easy-to-assemble robot (a benefit for swarm robotics, as developing robot swarms means making lots of robots).
- With each iteration, we make the robot more modular, and easier to waterproof.
- The current iteration is constructed using PETG and TPU as the primary structural materials, selected for their rigidity and flexibility respectively.
- The robot is actuated by two 9g servo motos and a small gear DC motor driven by an L293D motor driver.
- A syringe mechanism is incorporated for buoyancy. The electronics are controlled by an Arduino Nano, and the assembly is held together using screws.

Low Cost Robotic Development

The current robot has gone through several iterations, here are a few highlights.

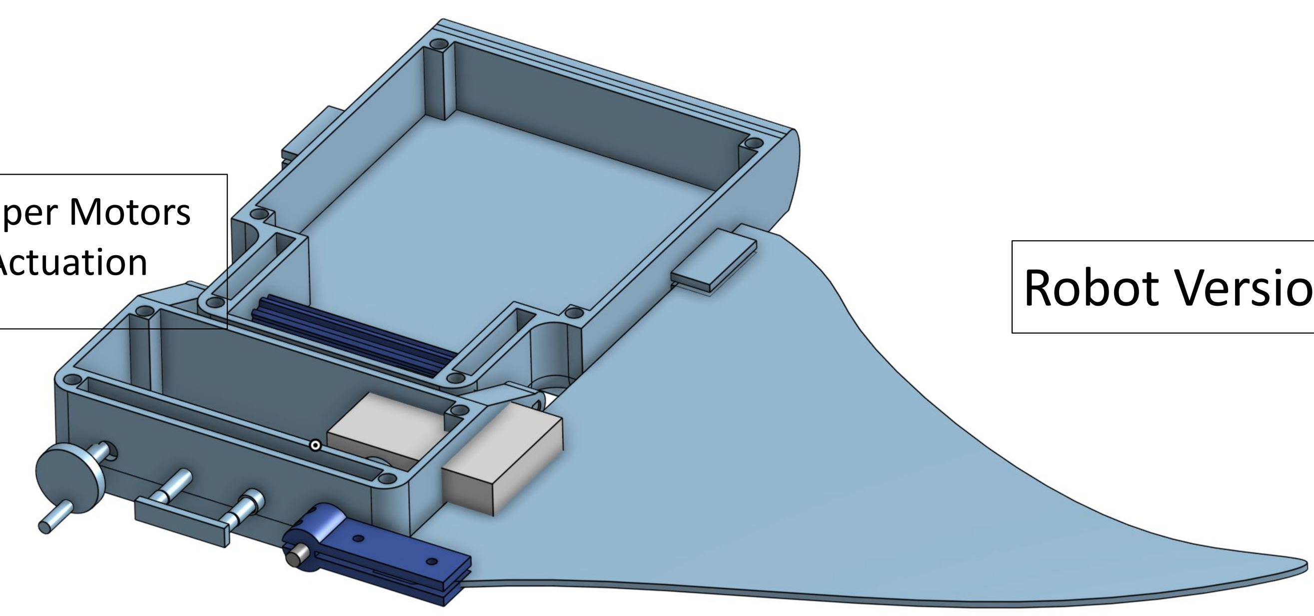
Stepper Motors for Actuation



Robot Version 1

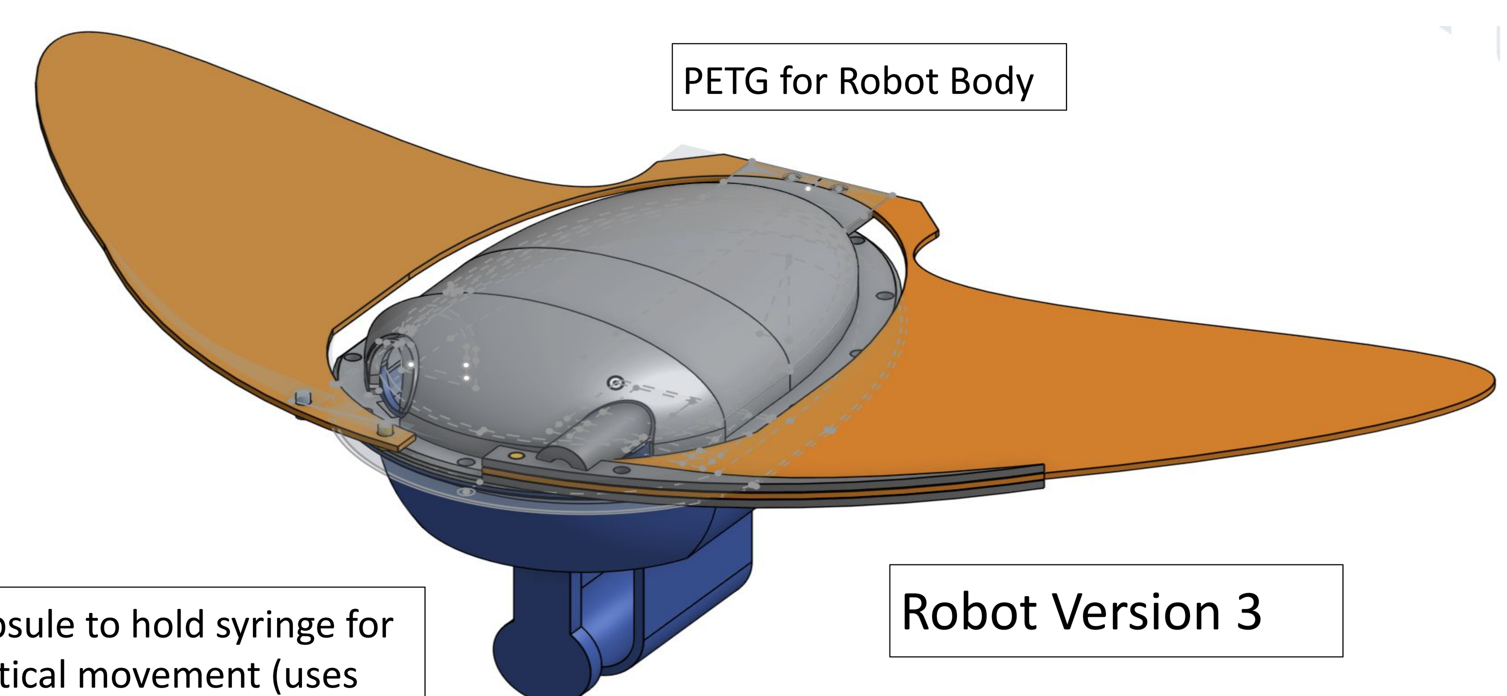
Central Gear for Vertical Movement

Stepper Motors for Actuation



Robot Version 2

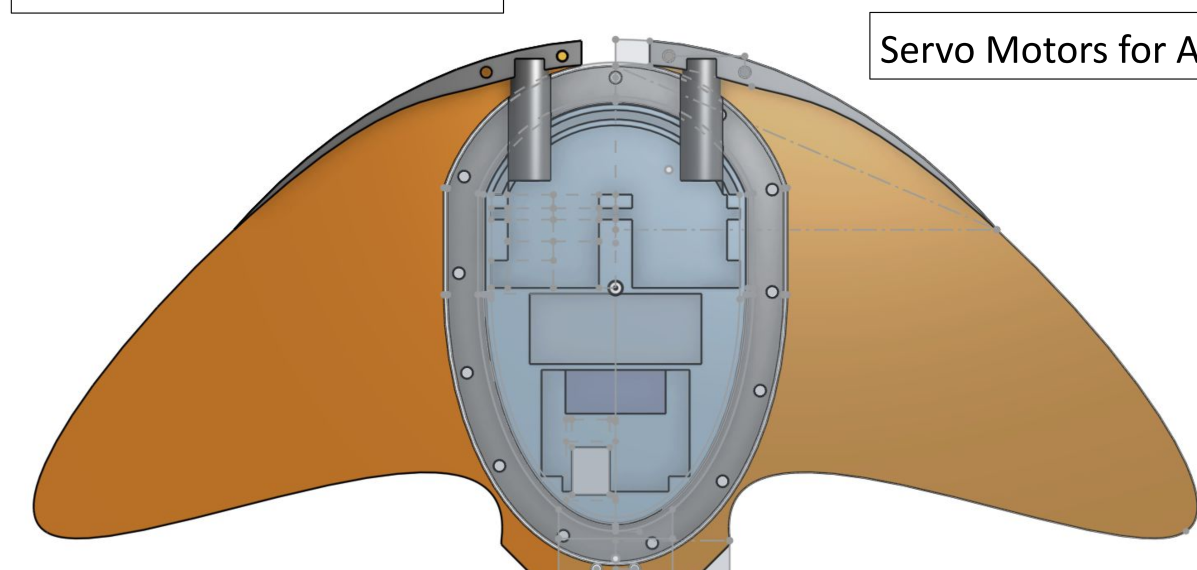
PETG for Robot Body



Robot Version 3

Capsule to hold syringe for vertical movement (uses geared DC motor)

Servo Motors for Actuation



Fins Made of TPU

Gasket for Waterproofing

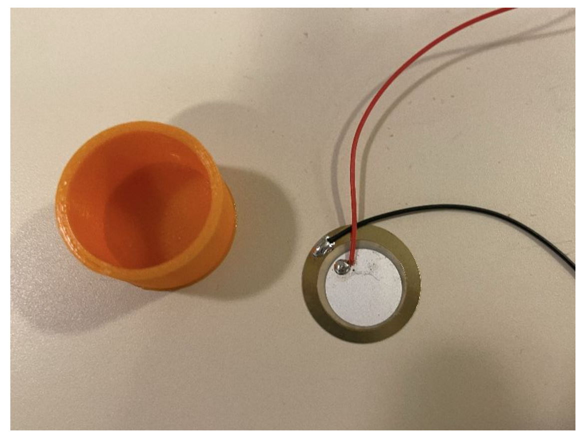
We are currently in the market for robot names for this newest iteration of robot. We could greatly appreciate any name suggestions.

PROPOSED COMMUNICATION SYSTEM

- Acoustics are the gold standard in underwater communicate, unfortunately they are also extremely expensive. We seek to develop cheaper sensors and effective communication protocols for this project

Sensor Development

- We will be using cheap (~$1.66) 10mm piezoelectric disks as our sensing medium. Piezoelectrics are able to convert vibration energy from underwater sound waves to voltage, and when charged with a voltage, can create vibrations.
- Each robot will have two sensors: a buzzer, and a receiver. The buzzer will send signals and the reciever will receive them.

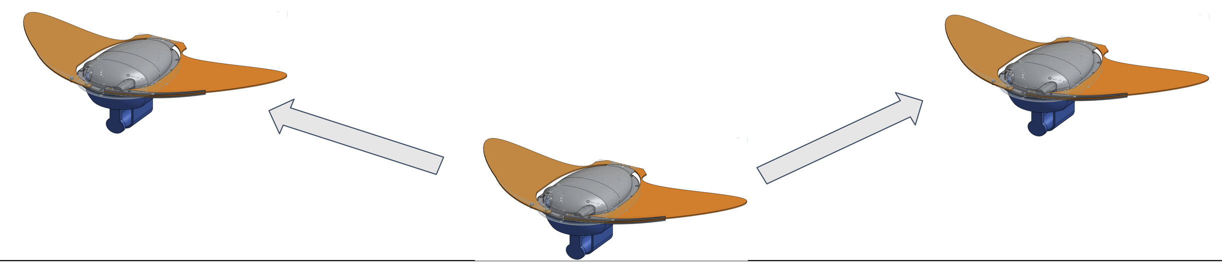


piezoelectric disk

Motor Casing

Communication Algorithm

- One time segment will be divided into n equal intervals, where n is equal to the number of robots in the swarm
- the robot will broadcast data (in a bit sequence) during its interval
- the data broadcasted will be proposed positions of the robot itself, (and maybe the proposed positions of neighboring robots, as well as the depth data



FUTURE DEVELOPMENTS

- Integration of additional sensors such as water quality monitors, temperature probes, and salinity sensors to expand the environmental monitoring capabilites of the swarm.
- Deployment in open-water environments to evaluate performance beyond controlled testing conditions.
- Scaling the swarm to a larger number of units to access communication reliability and protocol robustness at scale.
- Exploration of military and low-cost UAV data collection applications.

REFERENCES

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